CS60050 - Machine Learning

Assignment 2

Song Popularity using Neural Networks

Bratin Mondal

21CS10016

Department of Computer Science and Engineering,
Indian Institute of Technology Kharagpur

1 Problem Statement

1.1 Objective:

This project aims to build neural network models for song popularity prediction based on a comprehensive set of music characteristics.

1.2 Model Specifications:

1. **ANN Specification 1**
    - No of hidden layers: 1
    - No. of neurons in hidden layer: 32
    - Activation function in the hidden layer: Sigmoid
    - 1 neuron in the output layer.
    - Activation function in the output layer: Linear
    - Optimization algorithm: Mini Batch Stochastic Gradient Descent (SGD)
    - Loss function: Mean Squared Error (MSE)
    - Learning rate: 0.01
    - No. of epochs = 200
2. **ANN Specification 2**
    - No of hidden layers: 2
    - No. of neurons in the 1st hidden layer: 64
    - No. of neurons in the 2nd hidden layer: 32
    - Activation function in both the hidden layers: ReLU
    - 1 neuron in the output layer.
    - Activation function in the output layer: Linear
    - Optimization algorithm: Mini Batch Stochastic Gradient Descent (SGD)
    - Loss function: Mean Squared Error (MSE)
    - Learning rate: 0.01
    - No. of epochs = 200

2 Modules

2.1 Preprocess

Purpose: Preprocesses the input dataset by dropping duplicates, normalizing the data, and splitting it into training and testing datasets.

2.2 DataLoader

Purpose: Loads the training and testing datasets and splits them into batches of size 32 for efficient training.

2.3 WeightInitialiser

Purpose: Initializes the weights of the dense layers in the neural network. It generates random weights within a specified range.

2.4 BiasInitialiser

Purpose: Initializes the biases of the dense layers in the neural network. It initializes biases for each layer to zero.

2.5 ForwardPass

Purpose: Performs the forward pass of the neural network. It calculates the outputs of each layer using the given input data and weights.

2.6 BackPropagation

Purpose: Performs the backward propagation of the neural network. It updates the weights and biases of the network based on the calculated gradients.

2.7 Training

Purpose: Trains the neural network by iterating over epochs. It utilizes forward pass and backpropagation to update model parameters and computes accuracy for each epoch.

2.8 Predict

Purpose: Predicts the labels for the training and testing datasets using the trained neural network model. It utilizes forward pass to obtain predictions.

2.9 curAccu

Purpose: Calculates the accuracy of the model by comparing predicted labels with the ground truth labels.

2.10 BatchSplit

Purpose: Splits the dataset into batches of size 32 to facilitate mini-batch training.

2.11 gradSig

Purpose: Calculates the gradient of the sigmoid function. It's used in backpropagation for layers with sigmoid activation.

2.12 sig

Purpose: Calculates the sigmoid function. It's used as the activation function for the hidden layers.

2.13 reluDerivative

Purpose: Calculates the gradient of the ReLU function. It's used in backpropagation for layers with ReLU activation.

2.14 relu

Purpose: Calculates the ReLU function. It's used as the activation function for the hidden layers.

3